

Data Protection and Recovery

(45 theory lessons, 30 practice lessons)



Chapter 3:

Logical damage on the file system

& How To Fix Them

(45 theory lessons, 30 practice lessons)



Contents

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Common reasons

Common reasons

Malicious programs:

- There are many programs written for the purpose of data destruction:
 - Complete destruction (e.g. disk formatting).
 - Encrypting the data and demanding a ransom for data decryption and restoration.
 - Sometimes the data is completely damaged, even if the victim accepts the payment, it is still not possible to recover the lost data.

Common reasons

Buggy programs:

- There are plenty of imperfectly written programs that result in unexpected consequences
 - Buggy performance and create wrong data
 - Writing correct data in the wrong place making existing data faulty
 - Unintentional mass deletion of data
 - Formatting a wrong volume
 - ...

Common reasons

Hardware damage:

- Unstable storage devices can cause data corruption.
- For example, if an electric shock occurs while the data writing operation is in progress, the originally intended place of writing will be shifted and the old data will be overwritten.

Common reasons

User confusion:

- Users themselves cause unintended data corruption:
 - Data deletion/format/ghost due to carelessness or some unintentional mistakes (e.g., accidentally touching the delete/enter key on the keyboard).
 - Sometimes the operation could be performed exactly, but it takes a while to realize that some deleted data needs to be restored.

Common reasons

Intentional destruction:

- Some data corruption could be the consequence of unhealthy competition or personal conflict.
- One of the targeted objectives is computer data and the damage caused can be enormous.
- Although it may take a long time to build data, deleting or formatting the whole volume can be very quick. In these cases are said to be impossible to recover the data, however, in fact, it can sometimes be.

Specific situations and solutions

Wrong description of Partition:

- HDD, SSD, USB, Memory card, or generally large storage devices except optical discs can hold many Partitions. For a variety of reasons, the table that describes the location and size of the Partitions on the disk can be wrong, as a result, the system can not recognize the disks (volumes) and leading to the loss of all data on all volumes.
- The Partition table is located in the Master Boot Record (MBR) and is extremely important so there is usually at least one copy in another sector. Specifically, the table located at offset 1BE on the MBR has a size of 40h bytes, and contains 4 entries (means that each entry has a size of 10h bytes). After that table is the last 2 bytes of the MBR which are also used to identify the BR (always 55, AA).

Specific situations and solutions

Wrong description of Partition:

- To fix the error of wrong information in the Partition table, we will detect its correct copy and copy it back to the MBR.
- There are many sectors on a disk, however, copies of the MBR are usually located in the first and last sectors of the disk. Therefore, it is possible to limit the search range to only a few thousand sectors, and if unsuccessful, then expand to the entire disk. To detect the copy of the MBR, we start with the identification sign (last 2 bytes), then check the first 4 bytes of each entry to see if it is 00 or 80h. If it's true, then check the rationality of position and size of each entry (the total does not exceed the disk space). After finding out the right copies, then try to put them in the MBR in turn for validity test.
- In case you can't find the copies out, try to check the location of the BR (Boot Sector - the first sector of the volume) to know the location and size of each volume, and then rebuild the Partition content table.

Specific situations and solutions

Wrong parameter of Volume:

- Each volume must have a format architecture for the filesystem to store on. The information about the location/size of important components in the format architecture is organized right in the Boot Record (BR). If there is an error with only 1 parameter, the volume will often be misunderstood by the OS and cannot retrieve the data. In this situation, the more you try to use it, the more damage it can cause.
- BR appears to be greatly important and it is often necessary to have several backups of it. Plus, copies of BR are usually at the top of the volume, so finding it out doesn't take much time. However, the right restoration way of BR data is not about to search for the correct copy of BR and copy it back at the first step. Such a search will lead to multiple results and may include incorrect results (for sectors that are not BRs or BRs that are not of the volume in question). In fact, even in the BR itself the information about its copy is already provided, we just need to check on the right place to get it and further verify its precision. If it is not valid, then proceed with the search.

Specific situations and solutions

Wrong parameter of Volume:

- The identification sign of BR is also located in the last 2 bytes like the MBR, but still, we have to check a bunch of other information for authentication purpose. After the copies are found out, put them in BR in turn for validity test.
- If not found, the content of the relevant area can be consulted to figure out the position and size of each element, based on their identification signs as well as common default values (e.g. the cluster size is usually 8 sectors, the number of FAT tables is usually 2, the root directory size is usually 512 entries or the starting cluster is 2, ...), and then rebuild the volume parameter table. Reconstructing this content can be quite simple if the number of wrong parameters is small: we can refer to the values of other parameters to point out the appropriate values for the wrong ones.

Specific situations and solutions

Wrong folder table and Cluster table:

- The cluster table in the form of a linked list combined with clustered index usually reveal some logical errors. However, because the system is often organized into 2 parts, they can be used to make up for each other and fix the error (this is done by OS).
- In case the folder table is logically wrong while the cluster table is correct, the file content can still be restored (but the name and attributes may be gone), and it is not considered to cause a great loss.
- The situation that all tables are wrong at the same time usually occurs when the disk is formatted or ghosted. Since descriptive information about attributes and file content has been lost, recovery can not be guaranteed without luck. When the disk is in this state, you should not copy anything to it. The more data you put in the disk, the more old content will be lost. If possible, create an image of the volume as soon as possible (now that recovery can be done on the image itself, without access to the volume).

Specific situations and solutions

Wrong folder table and Cluster table:

- The basis for the data recovery in this case according to FAT architecture is that the sub-folder tables stored in the Data area can be mostly intact, and the file content is usually less fragmented. Hence, with the use of folder entries we can figure out the name and attributes of the file (first 2 entries "." & ".." can also be used to extract a lot of information), and 2 important attributes used to define the file content are <start cluster> and <size>. Speaking of the fragmented files (accounting for a very small percentage), although we need to use artificial intelligence algorithms, the recovery is definitely very tough and the success rate is difficult to reach 100%.
- On NTFS-based partitions, the old system information will be lost and the MFT table will be changed after formatting, but the cluster size usually remains 8 and some old record files still exist and are detectable. When the information about the MFT location is no longer available, it must be searched on each cluster to find it. If \$MFT record file in the MFT table is gone, the \$MFT record file can still be found on the \$MFTMirror file.

Specific situations and solutions

Deleted files/ folders:

- Because the common deletion mechanism is to compress the data and put it in the Recycle Bin, then delete it from there, let's check in the bin first to see if the deleted data is still there. if so, then the recovery is very simple and fast. This is a feature that enhances data safety for users designed for the system.
- In recent years, cloud computing has developed strongly and online data storage - backup services with real-time synchronization capabilities are welcomed and used by many people (e.g. dropbox, google drive, one drive, etc.). When the data on the computer is irrecoverably deleted (as in the case of vandals using specialized data deletion software) but has been synchronized on the network disk, it is only necessary to use the corresponding support in the services to retrieve data, can even be traced to know exactly when and how the operation that destroyed data was performed.

Specific situations and solutions

Deleted files/ folders:

- If both of the above cases can not be done, it is possible to use the entry corresponding to the data to be recovered, thereby determining the starting cluster index and the size of the file (because when deleting, the system only marks to the entry, not delete all the content). Then check the cluster table to find the consecutive empty cluster sequence corresponding to the cluster number of the file. It is very possible that the file content corresponds to this cluster range. In case there are some incorrect clusters, remove them out of the sequence and get the next empty cluster.

Specific situations and solutions

Data with lost description:

- The most complicated case: the descriptive information about the data has been completely lost (e.g. being overwritten). Because the file name and attributes no longer exist, we must clearly determine in what format the file needs to be restored and corresponding content must have a distinctive identification sign. Depending on the current volume status, we will know which sectors or clusters to scan:
 - For the volume that has not been changed in format (since the data is lost): detection is performed on empty clusters.
 - For volume that has been formatted after data loss: detection is performed on clusters or sectors of the old system, where these clusters / sectors are extracted from existing empty clusters.
 - With the volume corrupted on the formatted structure (no longer accessible at the operating system level): try to determine the location and size of the cluster and search on the clusters. Searching on sectors is performed only when the cluster can not be identified which takes much time and less accurate

Specific situations and solutions

Data with lost description:

- On the basis of locating the clusters, we can read the clusters corresponding to the folder table, from which we can detect the entry corresponding to the file needs to be stored, get the starting cluster information and the file size, then file restoration can be done.

Experiments and exercises

Experiments

Exercises

Questions and Answers